

Emergency Dispatch Priority and Coordination System

Final Project Report

ENSE707 Software Quality Assurance

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1. Project Completion and Final Implementation

- ☐ *Complete the software solution or quality improvement prototype proposed in Assessment 1*
- ☐ *Implement the key functional requirements and acceptance criteria*
- ☐ *Document and justify any changes to scope, requirements, architecture, technology, or implementation*
- ☐ *Demonstrate the final system through a working prototype*
- ☐ *Provide a GUI where user interaction is required, or justify an alternative interface (API, CLI, backend service, automated testing framework, or QA dashboard)*

1.1 Project Overview

There is a high volume of calls received by emergency dispatchers every day in New Zealand alone. Sometimes there are too many cases that office workers have to go out into the field to keep up with all of them (*1News Reporters, 2026*). This causes stress on the current emergency system, which underlines the need for a new response system that can effectively manage all emergency cases and coordinate response units for these cases. The Emergency Dispatch Priority and Coordination System was developed to improve the management of emergency calls, case handling, prioritisation, department routing, and response-unit coordination. The prototype from phase I allows the dispatcher to create a case and put it through the system, which assigns a priority. The case is then routed to the assigned departments, where they can view the case details on their dashboards with very quick updates. After a unit has finished a case, they can update it on their dashboard, and once all units have signed off, the case closes.



1.2 Final Scope and Changes Since Assessment 1

The original scope from assignment 1 was to develop an emergency dispatch priority and coordination system prototype that managed emergency cases, case prioritisation, case routing to departments, and response-unit coordination. The scope for the final assignment includes some changes to the features due to new findings, corrections required from the previous scope, as well as new ideas.

Scope Area	Assessment 1 Scope	Final Assignment Scope	Justification
Case recording	Dispatchers can record caller details, incident type, description, location, severity, and required departments in a form. Submitting this form creates a new case.	No scope change. The required inputs remain validated, and the case number and timestamp are generated automatically during case creation.	Case recording provides the information that is used in prioritisation, routing, unit assignment, and testing. The automatic timestamp and case number improve consistency.
Prioritisation	The system automatically determines the case priority using the department-related rules. The dispatcher's selected severity is used when there are no keyword matches.	Scope changed. An authorised dispatcher can override the calculated priority by giving a mandatory reason, which is stored in the audit log.	Priority override allows a miscalculated priority to be corrected. The original priority, updated priority, user, reason, and the timestamp should all be recorded for accountability,
Routing	Cases are routed to every department selected by the dispatcher. Multi-department cases appear in all of the selected departments' interfaces.	No scope change. Selected departments can view the case on their dashboards. Cases are excluded from unrelated departments' dashboards.	This workflow satisfies the multi-department routing scope from assignment 1. There is no external emergency-service integration in this prototype.
Unit assignment and coordination	The first available unit from each required department will be assigned to the case. If no compatible units are available, the case will be in a queue.	Scope changed. Unit assignment assigns the nearest available unit using its stored location data. If no unit is available, the case remains in the queue.	Nearest-unit selection can reduce travel distance for the selected response unit and the expected time of arrival. This is supported by location data that

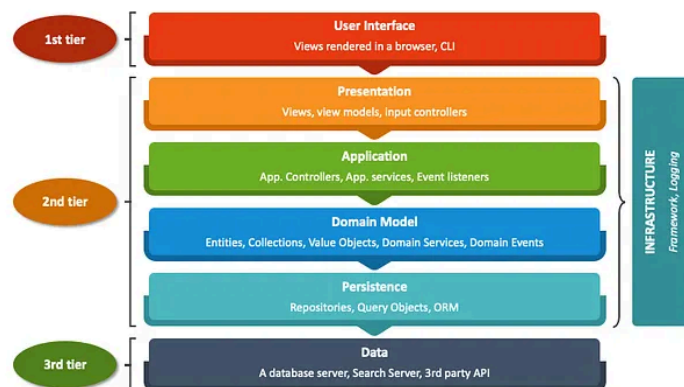
			is measurable, defined rules, and repeatable tests.
Case lifecycle, sign-off, and history	Cases use the following states: Open, In Progress, Closed. Each unit signs off individually, and the case only closes when all units have signed off. Dispatchers can search previous cases by caller name, case number, or date.	No scope change. The case lifecycle, unit sign-off, and case history are kept the same. Case closes after all required department response units have signed off. History remains searchable by the inputs mentioned.	The lifecycle prevents cases from being closed too early on multi-department cases, and keeps all states consistent throughout the case lifecycle. The history supports traceability and access to previous cases.
Role-based GUI	There are separate interfaces and restrictions provided for dispatchers, departments, response units, and the prototype admin role. Department and response unit users are restricted to their assigned scope.	No scope change. The role-based interfaces and server-side access controls are kept. The admin is prototype only and is used for testing and demonstration.	Role-specific interfaces reduce the complexity and prevent users from accessing pages they shouldn't be allowed to access. Admin is kept for the prototype for easier testing and demonstration.
Architecture & Technology	This system uses C#, .NET, Razor Pages, HTML/CSS, MSTest, and GitHub with a modular design.	No scope change. The structure, repository abstractions, priority strategy, dependency injection, and central dispatch service are kept.	Keeping the existing technology stack avoids migration and regression risks. Explaining the architecture shows how the prototype supports maintainability, extensibility, and testing.
Security and Persistence	The system uses in-memory repositories and fake accounts in text files. Operational data was lost after restarting the program. Role-based authorisation was	Scope changed. Includes permanent database storage, password hashing, encryption at rest, and a persistent audit log. The audit log records important actions related to a case,	Persistent storage prevents important data from being lost after the program restarts. Password hashing and encryption reduce the risk of sensitive data being exposed if

	implemented, but not permanent storage or encryption.	timestamps, and the responsible user.	the database is accessed directly. Audit logging improves accountability, traceability, and security review.
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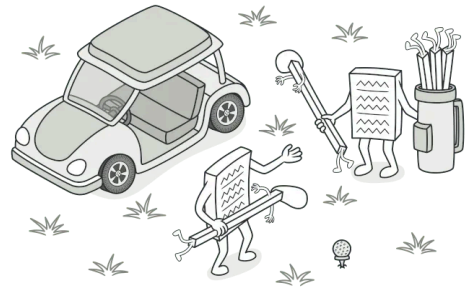
1.3 Final System Architecture and Design

There are six layers used in our system architecture: Domain, Application, Logic, Infrastructure, Web, and Authentication. This separation in the system means that each part can focus on one responsibility, while still contributing a part to the complete dispatch workflow.

The **domain layer** contains the system's main objects, with each of them controlling its own rules, functionality, and state changes in the system. For example, assigning cases, releasing a unit, or closing a case. The main objects are cases, units, departments, priorities, and case states. The **application layer** is the coordinator of the system and tells parts of the system "what needs to happen". DispatchService is an example in the application layer, where it coordinates the case creation, prioritisation, routing, and case-related tasks. The **logic layer** contains the rules about the system objects that are independent of the interface and database. For example, KeywordSeverityPriority.cs implements the priority logic in the system, where its rules can be changed independently. The **infrastructure layer** contains all of the department, case, notification and other repositories. These have direct access to the database and its chosen operations. The **web layer** uses ASP.NET Core Razor Pages to display the interface and pages of the system. This includes the sign-in page, dispatcher dashboard, department dashboard, and all other system pages. Lastly, the **authentication layer** holds files that are responsible for credential checking and making sure users can only access information that is permitted by their role.



Our system implements **two main design patterns: the repository pattern and the strategy pattern**. The repository design pattern gives an abstraction layer between the business logic and data storage. This gives a consistent way of managing data while hiding the implementation details (*GeeksforGeeks, 2026*). `ICaseRepository`, `IDepartmentRepository` and `IDispatchNotifier` all use the repository design pattern to separate the workflow from the storage. The strategy design pattern is a behavioural design pattern where a group of similar algorithms are used in a way that makes them interchangeable. This allows the algorithm to vary depending on what the client requires for their task, and enables these changes during runtime without modifying the existing code (*GeeksforGeeks, 2026*). In our system, the strategy pattern was used for case priority calculation. The `IPriorityStrategy` interface defines the framework to calculate priority, and `KeywordSeverityPriority` provides the implementation of that framework.



We used **dependency injection** to connect the different layers and components of our system architecture. All of the repositories, priority strategy, notification service, and the dispatch service are loaded when the application starts. Rather than every class creating its own dependencies created unnecessary load on the system, the components are provided to those classes instead. This reduces the coupling between components while also making testing easier because the repositories and services can be provided for testing instead of the real implementations.

The `DispatchService` was implemented as the **central service for the main system workflow**. It coordinates the case creation, priority calculation, storage, unit assignment, notifications, unit sign-off, and the reassignment of released units to the cases in queue. The intention of this service was to prevent similar workflows from being implemented in different locations of the system. The pages receive user input and display information, while `DispatchService` coordinates what should happen to the case and units (backend of the application).

Domain-controlled state was used to reduce the possibility of invalid states during runtime. The states of cases and response units are controlled by the domain objects instead of the user interface. The `Case` object controls whether the unit can be assigned, when a unit signs off, and whether the case

should be Open, Closed, or In Progress. The Unit object controls its availability and the current case assignment. With domain objects controlling their own states, it prevents invalid states. For example, an unavailable unit still being assigned to another case, or a multi-department case closing when not all units have signed off.

All of the design decisions and patterns implemented improve the quality of the system and make it easier to test and maintain. The **layered architecture helps with testing**, due to all the business logic being separated from the web interface. This allows the business logic to be tested without needing to load the interface, allowing for easier automated testing. The **repository design pattern reduces the coupling** between components because the application layer doesn't worry about how the data is stored. Overall, the design decisions make the system more modular and easier to modify without needing to also modify other unrelated layers. The persistent database, password hashing, and audit logging added improve the reliability, security and traceability of the system. This is because it prevents data loss after the program restarts, protected information is secure, and important actions on cases are recorded for accountability.

1.4 Final Implemented Functionality

These are the core functionalities that we have implemented during the first development of the initial prototype :

Requirements	Implemented Functionality	Acceptance criteria met	Evidence
FR-01 & FR-02	Emergency case recording and creation	The required caller, incident, location, severity, and department information is validated correctly. If valid, the submission creates a case with a unique timestamp and case number.	TC-01,TC-02, TC03, HTTP workflow
FR-03	Department unit management	Department users can only view their units and update unit location and their worker count. The unit availability is controlled by assignment and sign-off.	FR-03 tests, restart tests, role tests
FR-04	Automatic case	The system calculates priority using department rules. An	TC-04,

	prioritisation with authorised override	authorised dispatcher can override the priority result with a mandatory reason and, after entering the reason, the original priority, updated priority, user, and timestamp.	PriorityOverrideTests
FR-05	Department routing	The cases appear on every selected department's dashboard and are excluded from unrelated department dashboards.	TC-03, TC-05, TC-09, role tests
FR-06	Nearest available unit assignment and waiting queue	The nearest available response unit is selected using its stored location data. If no unit is available, the case remains waiting until a compatible unit is free.	TC-06 - TC-08, DistanceAssignmentTests
FR-07	Automatic dashboard updates	Each department dashboard only displays its relevant cases and includes new routed cases automatically. The page refreshes every 5 seconds.	TC-09, and rendered-page HTTP test
FR-08	Case status management	The case follows the defined state rules (Open, In Progress, and Closed) in waiting, assignment, partial sign-off, and completion.	TC-03, TC-06, TC-07, TC-10, database lifecycle
FR-09	Unit-availability management	Units become unavailable when assigned and available again after signing off. Two requests at the same time cannot give one unit two active assignments.	TC-03, TC-06 - TC-10, database lifecycle
FR-10	Assignment notifications	Each unit that is assigned receives a notification containing the case number, incident type, and location. If the notification fails, it should not undo successful case creation or assignment.	TC-03; notification-failure and restart test
FR-11	Response sign-off and closure	Units sign off individually; if units are still assigned to a multi-department case, its state should be In Progress. The case closes after all assigned cases have signed off.	TC-10, database lifecycle, HTTP workflow
FR-12	Case-history search	Dispatchers can search by caller name, case number, or date. Cases that match any of the fields are returned.	TC-11 and repository tests
FR-13	Persistent audit	Important actions include case creation, priority override,	AuditLoggingTests,

	logging	routing, assignment, sign-off, and closure. These are stored with the case action, user or system actor, and timestamp. Audit records should remain accessible after the program restarts.	override and lifecycle tests.
NFR-SO2 & NFR-U01	Role-based access and interfaces	Dispatchers, departments, and response units can access only their allocated pages and scope. Any unauthorised page requests should be denied.	TC-12, report access tests, HTTP tests
NFR-RO3	Permanent database persistence	Cases, units, assignments, notifications, accounts, and audit records should remain accessible after the application has restarted. Stored data should be retrieved without missing or changed fields.	SqlitePersistenceTests, database lifecycle
NFR-SO1	Encryption at rest	Sensitive account data should not be stored as readable plaintext in the local database. Authorised operations can encrypt, store, retrieve, and decrypt the data correctly.	SecurityTests
NFR-SO3	Secure password storage	Account passwords should be stored using password hashing. Correct passwords should be accepted, while incorrect passwords and direct hash comparison attempts are rejected.	SecurityTests
NFR-Po1	Dispatch completes within two seconds	Cases should be sent to the relevant department dashboards in under 2 seconds.	Passed on test machine

1.5 Final Prototype Evidence

The screenshot shows the 'Record Emergency' form in the Emergency Dispatch system. The form is divided into three main sections: Caller, Incident, and Response Required. The 'Caller' section has fields for 'Full name' and 'Phone number'. The 'Incident' section has fields for 'Incident type' (e.g., vehicle collision), 'Incident location' (Street address or clear location), 'Incident description' (Describe the situation, hazards, people involved, and relevant observations), 'Latitude' (-36.8485), 'Longitude' (174.7633), and 'Initial severity assessment' (Medium). The 'Response Required' section has checkboxes for 'Medical' (Ambulance and clinical response), 'Police' (Police response and scene safety), and 'Fire' (Fire and rescue response). At the bottom, there are 'Cancel' and 'Create Case' buttons.

Figure 1.5.1: Dispatcher case-recording form

This is the /Cases/Create page where the caller details, incident details, location, reported severity, and department selection are captured. It provides dispatchers with a simple GUI to enter details rather than the older, confusing page. The location input is sent to the unit-assignment service, where it checks the coordinates of the location and finds the nearest available unit from each selected department. All mandatory information is checked using server-side validation, and the severity is checked using the priority system, which calculates the final priority for the case. The priority can also be overridden with a mandatory explanation of why it is changing. It will then be saved to the audit log, and the case details will be saved on the dispatcher's dashboard (Figure 1.5.2).

The screenshot shows the Dispatcher Dashboard. At the top, a green notification bar states: "CASE-0DCB8611 created successfully. Routing and available-unit assignment are complete." Below this, the dashboard is titled "Dispatcher Dashboard" with a subtitle "Monitor current emergency cases and response progress." There is a "Live updates" indicator and a "Record Emergency" button. The main section is titled "NEEDS ATTENTION" and "Waiting for response units" with a note: "No compatible unit is currently available for one or more required responses." Below this is a table with columns: CASE, INCIDENT AND LOCATION, PRIORITY, REQUIRED RESPONSE, ASSIGNED, and CURRENT STATE.

CASE	INCIDENT AND LOCATION	PRIORITY	REQUIRED RESPONSE	ASSIGNED	CURRENT STATE
CASE-0DCB8611 09 Oct, 10:01	House fire 17 Queen street	High Override	Medical, Police, Fire	PCL-01, FIR-01	Waiting for Medical

Figure 1.5.2: Case on dispatcher's dashboard

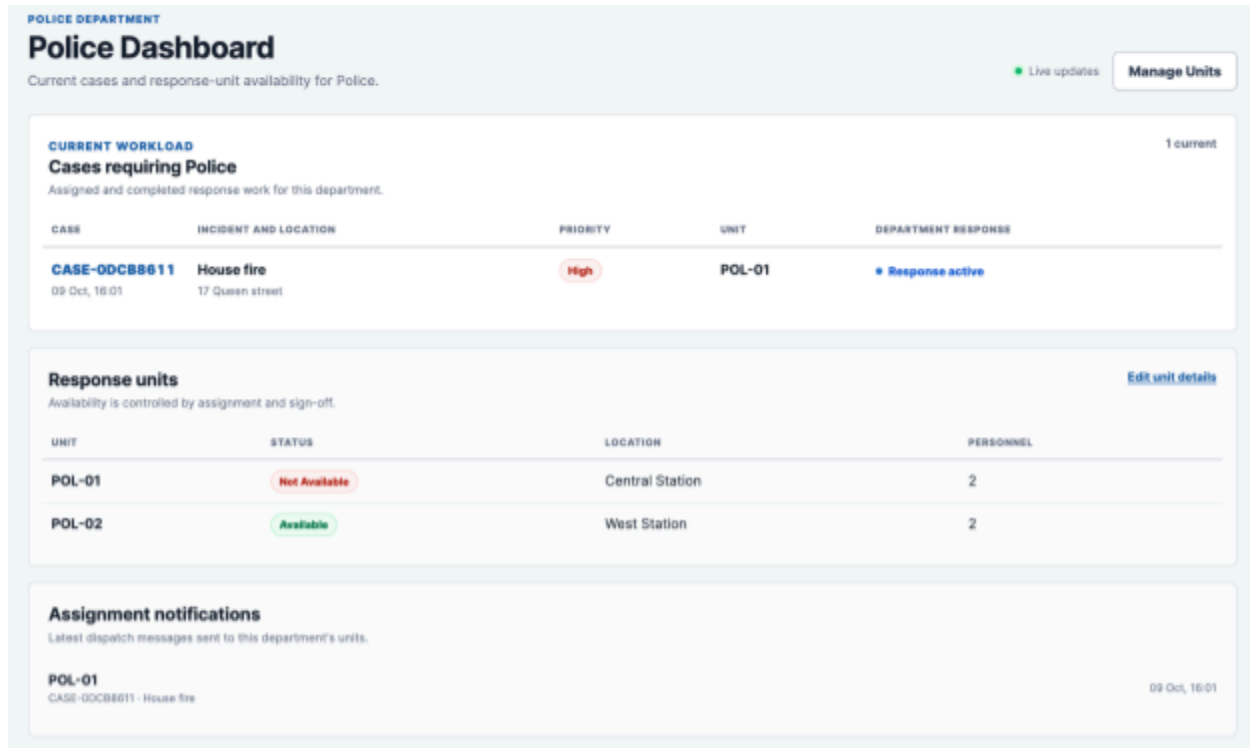


Figure 1.5.3: Department-specific dashboard

Figure 1.5.3 shows the police department's dashboard, where only cases assigned to this department will be displayed. Each dashboard shows the current cases that require police units, the list of all response units along with their availability status, and any notifications from the system. Each of these components is separated for easier viewing, with the goal of users being able to interpret critical information quickly. Dashboards reload every 5 seconds so that any new information is quickly updated, without the user having to manually reload the page, and reduce the chances of missed information.

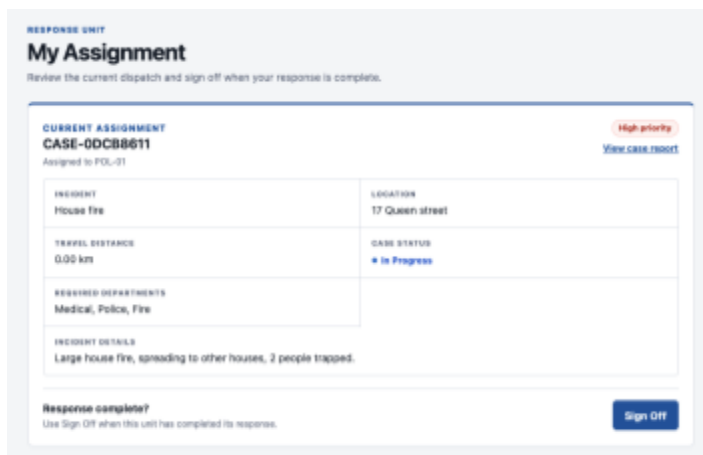


Figure 1.5.4: Response-unit workspace

Figure 1.5.4 shows the interface of the POLICE-1 unit, where they receive the current assignment and its details. The dispatcher's note is added because it provides important information that helps the unit prepare for the case. After the case is handled, the unit can sign off from the case, which only closes when all assigned units have signed off.

DISPATCHER RECORDS

Case History

Search previous emergency cases by caller, case ID, or date.

Caller name

e.g. Alex Morgan

Case ID

CASE-1234ABCD

Recorded date

dd / mm / yyyy

Search

A case is returned when any entered field matches.

All cases

2 found

Current and completed emergency records.

CASE	RECORDED	CALLER	INCIDENT AND LOCATION	PRIORITY	STATUS
CASE-D7CEA716 Medical	09 Oct 2026 16:51	Jane Doe 0219876543	Vehicle collision 385 Great South road	Medium	Open
CASE-ODCB88611 Medical, Police, Fire	09 Oct 2026 16:01	John Doe 0211234567	House fire 17 Queen street	High	In Progress

Figure 1.5.5: Case history showing case in queue

When no response units are available, the newly created case goes into a queue, with the status “Open”. Figure 1.5.5 shows the case that was just created with the status in the case history log. In the case history page, the dispatcher can search for cases by caller name, case ID, or the date it was created.

PROTOTYPE ADMINISTRATION

Persistent Audit Log

Important system actions are shown newest first.

TIME	ACTION	USER	CASE	UNIT	CHANGE	REASON
09 Oct 2026 16:51:32	Case created	admin	CASE-D7CEA716	—	— → Priority: Medium; Departments: Medical	Emergency case submitted.
09 Oct 2026 16:01:03	Unit assigned	admin	CASE-ODCB88611	FIR-01	Available → Assigned	Closest available unit selected at 0.00 km.
09 Oct 2026 16:01:03	Unit assigned	admin	CASE-ODCB88611	POL-01	Available → Assigned	Closest available unit selected at 0.00 km.
09 Oct 2026 16:01:03	Case created	admin	CASE-ODCB88611	—	— → Priority: High; Departments: Medical, Police, Fire	Emergency case submitted.

Figure 1.5.6: Audit log of case actions

Figure 1.5.6 shows the audit log of actions taken by different users. Every major action: creation, assignment, override, sign-off, and closure, is saved to this log to track who did what and to help with traceability. If an important event happened and someone had to figure out who performed a certain action, this log helps them find which user did it, making it easier for investigation if it occurs.

Case report CASE-D7CEA716: Emergency Dispatch

CASE REPORT
CASE-D7CEA716
 Recorded 09 October 2020 at 16:51

STATUS OPEN	CURRENT PRIORITY MEDIUM	REQUIRED RESPONSE Medical	COORDINATION Waiting for Medical
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Incident details
The information recorded when the case was created.

INCIDENT TYPE Vehicle collision	REPORTED SEVERITY Medium
DESCRIPTION Three vehicles in the crash, two moderately injured	
LOCATION 385 Great South road	
LATITUDE -36.848500	LONGITUDE 174.783300

Priority decision
Calculated and final priority values for this case.

CALCULATED PRIORITY Medium
FINAL PRIORITY Medium
OVERRIDE No manual override recorded

Caller details
Private information visible to dispatch and administration only.

CALLER NAME Jane Doe	CALLER PHONE 0219876543
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Response coordination
Assignment, distance, and sign-off evidence for each required department.

No response units have been assigned yet.

Waiting for response: No available Medical unit has been assigned.

Figure 1.5.7: Printable case report

All cases have a “print report” button which generates a report of the case details and its full lifecycle. It shows the incident details, priority decision including the override details if any, caller details, the response coordination, and finally the case activity (displayed like the audit log, but only actions linking to this case).

1.6 Remaining Implementation Limitations

The final prototype implements all of the important requirements of a real-life emergency dispatch system, but it is still a local hosted prototype. This requirements include case recording, department routing, prioritisation, response-unit assignment, and audit logging. The routing is done locally, so it does not communicate with anything outside of this local hosted prototype, like real life emergency services. The distance calculation required coordinates, and adding a coordinate finder was out of the

scope. Therefore, the dispatcher has to enter coordinates from a mock coordinate list, and use that for demos.

Dashboards are updated every five seconds using polling. In the future, we would like to add real-time push updates, which would allow departments to receive cases instantaneously, instead of five seconds later. Another limitation is that the SQLite database and locking has only been tested for this prototype, and is not viable for large distributed systems.

As an Emergency Dispatch Priority and Coordination system, the security is nowhere close to what it should be for a real life emergency system. Our prototype still lacks secure database management, authentication methods, and firewalls for system protection, so while it is a functioning prototype, it should not be applied to real life emergency scenarios or implemented into systems.

2. Requirements Verification and Traceability

- ☐ Review and finalise the functional and non-functional requirements from Assessment 1
- ☐ Update the Requirements Traceability Matrix
- ☐ Map requirements to implemented features, test cases, test results, and defects where applicable
- ☐ Identify any requirements that are partially implemented, unverified, or out of scope, and justify them

2.1 Finalised Requirements

From reviewing the 24 finalised requirements in the mid-term report, changes to some of the requirements would need to be made to increase the consistency, testability, clarity, and completeness. These are the requirements that need to be altered to consider the feedback provided in the mid-term report and further evaluation of each requirement:

Functional Requirements:

ID	Final Requirement
FR-01	The system shall provide a dispatcher interface for recording caller details, incident type, description, location, coordinates, reported severity and required response departments. The system shall record the submission timestamp.
FR-02	When valid emergency information is submitted, the system shall create one case containing the recorded details, a unique case identifier and a generated timestamp.
FR-03	When valid emergency information is submitted, the system shall create one case containing the recorded details, a unique case identifier and a generated timestamp.
FR-04	The system shall calculate the priority of a new case using the department-specific keyword rules applied to the case. If no keywords are matching, then the reported severity will be used. For multi-department cases, the highest result shall be used. An authorised dispatcher may override the priority after providing a good reason.
FR-05	The system shall route a case to every department selected by the dispatcher. The case shall be visible to the selected departments and hidden from unrelated departments.

FR-06	For each required department, the system shall assign the nearest compatible available unit using coordinates and distance. Unavailable units shall be excluded. If no compatible unit is available, the case shall remain waiting until one becomes available.
FR-07	The system shall provide each department with a dashboard containing only cases relevant to that department. The dashboard shall reload every five seconds so new information appears without having to manually refresh.
FR-08	A case shall be Open while waiting with no active assignment, In Progress while at least one required response is active, and Closed after every required response has signed off.
FR-09	When a unit is assigned, its availability shall change to unavailable. When it signs off, it shall become available and may be assigned to the oldest compatible waiting case.
FR-10	When a unit is assigned, the system shall create a notification containing the relevant case number, incident type and location. A notification failure shall not reverse successful case creation or case assignment.
FR-11	Each assigned response unit shall be able to sign off individually. A multi-department case shall remain In Progress until the final required response signs off, after which it shall close automatically.
FR-12	The dispatcher shall be able to search case history by caller name, case number or date. When several search values are provided, cases matching at least one entered value shall be returned.
FR-13	The system shall persist audit events for important actions, including case creation, routing selections, assignment, unit updates, priority overrides, sign-off and closure. Each event shall include the action, actor, timestamp and relevant case or unit details.

Non-Functional Requirements:

ID	Final Requirement
NFR-PO1	A locally submitted case shall be stored, routed and assigned to all relevant departments within two seconds under the prototype test conditions.

NFR-PO2	The system shall support at least 100 concurrent users while maintaining a response time of two seconds or less when navigating or refreshing an interface.
NFR-MO1	The system shall use separated domain, application, logic, infrastructure, authentication and web responsibilities. A failure in the non-critical notification component shall not undo successful case storage or unit assignment.
NFR-MO2	New roles and departments should be introducible without changing the existing core rules for case creation, routing, assignment and lifecycle status.
NFR-RO1	Case information made visible to each selected department shall remain complete and consistent with the case submitted by the dispatcher.
NFR-RO2	The deployed system should maintain at least 99.9% availability over a seven-day period, excluding planned maintenance.
NFR-RO3	Cases, units, assignments, notifications, accounts and audit events shall remain retrievable without missing or changed fields after the application and database are restarted.
NFR-RO4	The system shall continue accepting, displaying and updating cases without crashing when 100 cases are simultaneously open.
NFR-SO1	Selected sensitive account, caller, incident and location values stored in SQLite shall be encrypted using AES-256-GCM. A production deployment shall use TLS 1.2 or higher for information transmitted between users and the server.
NFR-SO2	Each authenticated user shall be restricted to the pages, actions and information permitted for their assigned role and department or unit scope. Direct requests to unauthorised pages or records shall be denied.
NFR-SO3	Account passwords shall be stored as salted PBKDF2 hashes rather than plaintext. Correct passwords shall be accepted and incorrect passwords shall be rejected.
NFR-UO1	The system shall provide tailored interfaces for Dispatcher, Department and Response Unit users, displaying only the tools and information required for their roles. The Admin interface is a prototype-only testing exception.
NFR-UO2	User-facing validation and error messages shall use understandable language, identify

	the problem where appropriate, and shall not expose exceptions, stack traces or debugging information.
NFR-03	After no more than one hour of training, a representative user shall be able to complete the defined core tasks for their assigned role without assistance.

The major functional changes were made to FR-03, FR-04, FR-06, FR-07, FR-08 and FR-11. For FR-03, it now separates the department unit information that can be edited from the availability, which is controlled by the system. For FR-04, it defines the department-specific keyword rules, the severity fallback, multi-departmental priority selection, and the new priority override behaviour. For FR-06, the system changes the first available unit to the closest compatible unit. In FR-07, the system defines the dashboards for each department and the 5 second polling. For FR-08 and FR-11, the system aligns with the lifecycle of individual units signing off, rather than the dispatcher controlling the case closure. FR-13 was new, which added the audit logging.

The non-functional requirements were also changed. NFR-03 requires the system to have permanent data storage that loads after the program restarts. NFR-S01 adds encryption at rest, and NFR-S03 adds password hashing.

The rest of the non-functional requirements and functional requirements were unchanged due to the ambiguity found in the previous improvement iteration of the requirements. All the finalised requirements are in **Appendix 5**.

2.2 Final Requirements Traceability Matrix

Functional Requirements Traceability Matrix:

ID	Final implementation	Verification Evidence	Result	Defect
FR-01	Validated Razor form records caller, incident, location, coordinates, severity and selected departments.	TC-01, TC-02 and HTTP workflow	Passed	D-06
FR-02	Creates one case with a generated identifier and timestamp. Initial status depends on unit availability.	TC-01, TC-03 and HTTP workflow	Passed	–
FR-03	Department users can view and update their units' location, coordinates and personnel count. Availability is controlled by assignment and sign-off.	FR-03 valid/invalid update tests, restart test and role tests	Passed	D-06
FR-04	Department-specific keyword rules calculate priority, with reported severity as the fallback. Authorised overrides require a reason.	TC-04 and PriorityOverrideTests	Passed	D-06
FR-05	Cases are visible to selected departments and excluded from unrelated department dashboards.	TC-03, TC-05, TC-09, role tests and HTTP tests	Passed for internal routing	–
FR-06	Selects the nearest compatible available unit using Haversine distance. Cases wait when no compatible unit is available.	TC-06–TC-08 and DistanceAssignmentTests	Passed	D-03

FR-07	Department dashboards contain only relevant cases and reload every five seconds.	TC-09 and rendered-page HTTP refresh test	Automated checks passed; visible timing requires manual confirmation	D-06
FR-08	Domain rules control Open, In Progress and Closed case states.	TC-03, TC-07, TC-10 and database lifecycle test	Passed	–
FR-09	Assignment makes units unavailable; sign-off releases them; locking prevents duplicate active assignment.	TC-06–TC-10, concurrency test and database lifecycle test	Passed	–
FR-10	Assigned units receive persistent case notifications. Notification failure does not undo a successful dispatch.	TC-03, notification-failure test and notification restart test	Passed	D-04
FR-11	Units sign off individually, and a case closes after the final required response signs off.	TC-10, database lifecycle test and HTTP workflow	Passed	–
FR-12	Dispatcher history supports OR-based search by caller, case number or date.	TC-11 and repository search tests	Passed	–
FR-13	Important case actions are stored persistently with actor, time, case and unit information.	AuditLoggingTests, PriorityOverrideTests and database lifecycle test	Passed	D-01, D-06

Non-Funtional Requirements Traceability Matrix:

ID	Final implementation or evaluation	Verification evidence	Result	Defect
NFR-PO 1	Database-backed multi-department dispatch must finish within two seconds locally.	TC-05; database-backed performance test	Passed on test machine: approximately 35 ms	–
NFR-PO 2	Support 100 concurrent users with responses within two seconds.	No representative concurrent-user load test	Unverified/out of scope	–
NFR-MO 1	Layered architecture and repository interfaces separate critical workflows from adapters; notification failure is isolated.	Architecture review; notification failure test	Partially verified	D-04
NFR-MO 2	Strategy and repository abstractions reduce changes to core workflow, but adding roles or department types still requires configuration and code changes.	Code and architecture review	Partially met	–
NFR-RO 1	Stored case information remains consistent across selected department projections.	TC-03, TC-05, TC-09; HTTP role tests	Passed for internal routing	–
NFR-RO 2	Maintain 99.9% weekly availability.	No deployed monitoring period	Unverified/out of scope	–
NFR-RO 3	Cases, units, assignments, notifications, accounts and audit events persist across restarts.	SqlitePersistenceTests; multi-restart lifecycle	Passed	D-01
NFR-RO 4	Continue operating with 100 simultaneously open cases.	No controlled 100-case operational or load test	Unverified	–

NFR-SO1	Selected sensitive database fields use AES-GCM; deployed TLS has not been independently verified.	Raw-database encryption and restart test	Partially passed	D-O2
NFR-SO2	Cookie authentication, middleware, role claims and scope checks restrict access.	TC-12; report access; real HTTP tests	Passed	D-O6
NFR-SO3	Passwords stored as salted PBKDF2 hashes; incorrect passwords rejected.	SecurityTests; restart authentication test	Passed	D-O2
NFR-UO1	Dispatcher, department, response-unit and prototype Admin interfaces are role-specific.	TC-12; page-model and HTTP tests	Functional access passed; usability remains manual	D-O6
NFR-UO2	Validation provides user-facing messages without exposing stack traces.	Invalid-input and login tests; manual interface review still required	Partially verified	D-O6
NFR-UO3	Users complete core tasks after no more than one hour of training.	No representative timed usability study	Unverified	–

2.3 Partially Implemented, Unverified and Out-of-Scope Requirements

3. Test Execution and Final Test Results

- ☐ *Execute the planned tests against the completed system*
- ☐ *Include relevant functional and non-functional tests (unit, integration, system, acceptance, regression, usability, security, performance, or accessibility)*
- ☐ *Include positive, negative, boundary, invalid-input, and error-handling scenarios where applicable*
- ☐ *Provide at least 15 final test cases with execution results*
- ☐ *Summarise the number of planned, executed, passed, failed, blocked, and not-run tests*
- ☐ *Explain failed tests, corrective actions, regression testing, and remaining quality risks*

3.1 Final Testing Scope and Approach

- Take from testing code

3.2 Final Test Cases

- All done take from testing code

3.3 Test Execution Results

- All done take form testing code

3.4 Failed Tests and Corrective Actions

3.5 Regression Testing

3.6 Remaining Testing Risks

4. Test Automation and Quality Engineering

- ☐ *Identify suitable test cases for automation and justify the selection*
- ☐ *Implement automated tests using an appropriate tool or framework where suitable*
- ☐ *Report automated test execution results using logs, screenshots, reports, or repository evidence*
- ☐ *Discuss the benefits, limitations, and maintainability of the automated test suite*
- ☐ *Where automation is not appropriate, justify the use of manual or exploratory testing*

4.1 Automation Selection

- Check project test files

4.2 Automated Test Implementation

- Check project test files

4.3 Automated Test Results

- Check project test files

4.4 Automation Maintainability

4.5 Manual and Exploratory Testing

5. Continuous Integration and DevOps QA

- ☐ *Describe the development and integration workflow*
- ☐ *Implement a simple CI workflow or equivalent automated quality check where appropriate*
- ☐ *Define quality gates (successful build, passing tests, linting, code review, or coverage threshold)*
- ☐ *Demonstrate smoke or sanity testing where relevant*
- ☐ *If CI/CD is not technically appropriate, justify an alternative quality workflow*

5.1 Development and Integration Workflow

5.2 Continuous Integration Pipeline

5.3 Quality Gates

5.4 Smoke and Sanity Testing

5.5 CI Evidence and Limitations

6. Defect Management and Root Cause Analysis

- ☐ *Maintain a final defect register covering defects found during development and testing*
- ☐ *Define severity, priority, and defect status categories*
- ☐ *Describe the defect lifecycle used in the project*
- ☐ *Analyse at least 3 significant defects using root cause analysis*
- ☐ *Propose corrective actions and defect-prevention measures*
- ☐ *Explain how defect analysis improved product or process quality*

6.1 Defect Management Approach

6.2 Severity, Priority and Status Definitions

6.3 Defect Lifecycle

6.4 Final Defect Register

6.5 Root Cause Analysis: Defect One

6.6 Root Cause Analysis: Defect Two

6.7 Root Cause Analysis: Defect Three

6.8 Corrective and Preventive Actions

7. Quality Testing

- ☐ *Conduct quality testing relevant to the project (security, performance, usability, accessibility, compatibility, reliability, or maintainability)*
- ☐ *Address at least 2 quality areas where applicable*
- ☐ *Use risk-based reasoning to justify the selected testing activities*
- ☐ *Document testing evidence, results, limitations, and recommended improvements*

7.1 Risk-Based Selection of Quality Areas

7.2 Quality Area One: [e.g. Security]

7.3 Quality Area Two: [e.g. Performance]

7.4 Results and Limitations

7.5 Recommended Improvements

8. Quality Metrics and Final Release Decision

- ☐ *Present relevant quality metrics (test pass rate, defect severity distribution, requirement coverage, test coverage, automation coverage, code quality indicators, commit activity)*
- ☐ *Evaluate the completed system against requirements, acceptance criteria, and quality attributes*
- ☐ *Summarise testing results, defect status, metrics, limitations, and residual risks*
- ☐ *State whether the system is ready for release, conditionally ready, or not ready for release*
- ☐ *Justify the release decision using evidence*

8.1 Final Quality Metrics

8.2 Requirements and Acceptance-Criteria Evaluation

8.3 Outstanding Defects and Residual Risks

8.4 Final Release Decision

9. AI-Assisted Development and Critical Reflection

- ☐ Explain how GitHub Copilot was used (implementation, testing, debugging, refactoring, or QA activities)
- ☐ Provide representative evidence of AI-assisted development
- ☐ Explain how AI-assisted outputs were reviewed, modified, tested, accepted, or rejected
- ☐ Identify quality risks found in AI-assisted outputs (incorrect logic, insecure code, poor maintainability, unsuitable assumptions)
- ☐ Reflect on how AI-assisted development affected productivity, learning, testing, software quality, and professional practice

9.1 AI-Assisted Activities

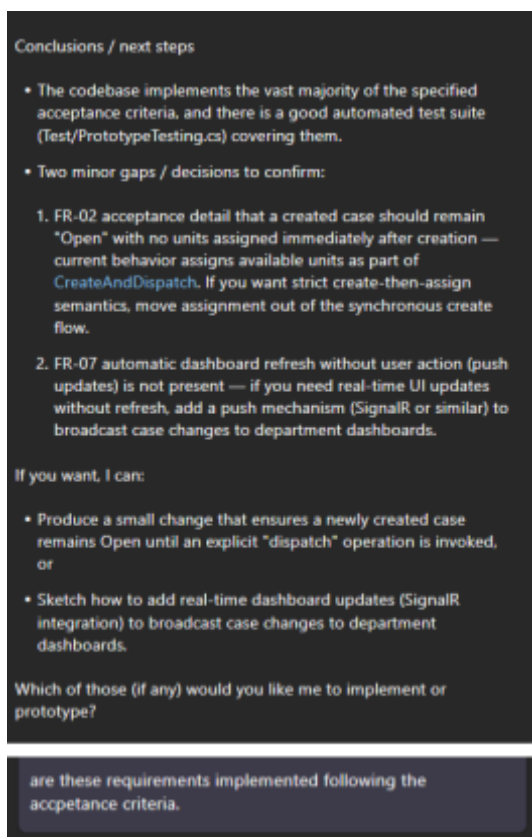


Figure 1.2.1: Github Copilot Evaluation

In the previous assignment, the functional requirements and non-functional requirements are defined and the acceptance criteria are created to describe expected behaviour of the core functional requirements. GitHub Copilot has been used to review the requirements that are gaps identified of core functionality with acceptance criteria. FR-02 and FR-07 are flagged as gaps that need to be considered for further changes. FR-01, FR-04, FR-05, FR-06, FR-08, FR-09 and FR-11 are not found as issues and are considered potential issues by GitHub Copilot. It identified that FR-02 acceptance criteria is not fully met, this criteria in FR-02 “Given a case is newly created from a recorded call, when creation completes, then the case status is set to “Open” with no units assigned to it yet”. FR-07 is flagged because GitHub described that the dashboard does not automatically refresh without user action, which doesn’t meet this: “Given a department’s dashboard is open, when the information of a case changes e.g. priority, location, then the dashboard should refresh and show the new changes automatically.” and “Given a department’s dashboard is open, when a case is routed to that department, then the dashboard should display the new case automatically.” When personally

investigating whether the FR-02 implementation meets the acceptance criteria gaps during testing, there isn't an issue because if there is a unit available, then the case would be assigned, thus receiving the status of "InProgress". The acceptance criteria is only met when there are no available units in that relevant department. Therefore, the acceptance criteria would need to be updated that the criteria is only for when there are no available units. For FR-07, after personal testing of the prototype, it has met its expected behaviour since the dashboard is updated without manually refreshing. The suggested GitHub flagged issues because it assumed that automatic updates mean adding a push mechanism (SignalR) to satisfy the stated criteria.

9.2 Representative GitHub Copilot Evidence

9.3 Human Review and Validation

9.4 AI-Related Quality Risks

9.5 Critical Reflection

10. Quality Improvement and Future Recommendations

- ☐ *Identify the main quality lessons learned from the project*
- ☐ *Recommend improvements to the development and QA process*
- ☐ *Propose future improvements to testing, automation, CI/CD, monitoring, security, usability, accessibility, or maintainability*
- ☐ *Explain how the project could evolve toward stronger continuous quality engineering practice*

10.1 Key Quality Lessons

10.2 Improvements to the Development and QA Process

10.3 Future Testing and Automation Improvements

10.4 Future Continuous Quality Engineering

11. Team Contribution and Professional Reflection

- ☐ *Provide a contribution statement for each group member*
- ☐ *Provide evidence from GitHub commits, project artefacts, testing work, documentation, and implementation tasks*
- ☐ *Include a brief individual reflection from each group member*

11.1 Contribution Statement

11.2 GitHub and Project Evidence

11.3 Individual Reflection – Member One

11.4 Individual Reflection – Member Two

11.5 Individual Reflection – Member Three

Delivarables (Notes)

Prepare a 15-slide presentation covering: title/members/problem/stakeholders/scope, key requirements, final prototype and changes since Assessment 1, testing approach/coverage/results, automation/CI/CD, defects and RCA, quality metrics and release decision, Copilot use and limitations, individual contributions and lessons learned

All students participate in the demonstration and show the working prototype

Demo includes key features, important test cases, automated tests/CI results, defect fixes, evidence major requirements are satisfied

Be prepared to answer questions on the project, code, testing, quality artefacts, AI-assisted development, and individual contributions

Deliverables

- Final Project and Software Quality Assurance Report: 6,000–7,000 words
- Source code and GitHub repository link, with evidence from GitHub commits
- Contribution statement for each group member
- Project presentation
- Turnitin Similarity Report

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